RUDRA CHAKRABORTY

BIOMEDICAL ENGINEER | AI, MACHINE LEARNING AND ROBOTICS
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EDUCATION

B.Sc. in Biomedical Engineering | CUET Apr 2022–Jun 2026
Chittagong University of Engineering & Technology | CGPA: 3.26/4.00.
HSC: Comilla Victoria Government College | 2020 | GPA: 5.00/5.00.
SSC: Nizmehar Model Pilot High School | 2018 | GPA: 5.00/5.00.
Relevant coursework: Machine Learning; Biomedical Image Processing; Biomedical Signal Processing; Embedded Systems and Interfacing; Biomedical Instrumentation.

TECHNICAL SKILLS

Programming: Python, MATLAB, C, TypeScript, React, Next.js.
AI and ML: CNN transfer learning, MDM/LDA, MediaPipe, OpenCV.
Robotics and sensing: Soft robotics, ESP32, Arduino, SolidWorks; sensor fabrication and calibration.
Tools and platforms: EEGLAB, Simulink, Proteus, Firebase, n8n, Blender; Codex, Cursor AI, Google Antigravity.

UNDERGRADUATE THESIS

Smart Actuator for Paralyze Patient Hand Rehabilitation
Supervisor and Advisor: Dr. Sumit Majumder, Assistant Professor, Department of Biomedical Engineering, CUET.
- Integrated MediaPipe hand tracking with ESP32 pneumatic actuation for independent finger control; developed four silicone-actuator design iterations in SolidWorks.
- Built MotionMirror (Next.js/Firebase) with doctor-patient WebRTC calls, therapy controls and 3D hand visualization.

RESEARCH AND ENGINEERING PROJECTS

DIY Flexible Sensors and Calibration Pipeline
- Fabricated silicone-encapsulated Velostat flex sensors. Developed a reusable five-channel Python/Arduino pipeline for per-sensor calibration of relative bending, with guided rest/bend capture, adaptive baselines, live plots and CSV logging.

EEG Motor Imagery Classification | ICCIT 2026 manuscript under review
- First-listed author; built MDM-LDA on public EEG data from 50 stroke patients using 449 covariance/coherence features. Reported 82% mean accuracy and kappa 0.62 in repeated within-subject splits.

Skin Cancer Classification Using Deep Learning | EfficientNet-B2, HAM10000
- Fine-tuned a seven-class CNN using lesion-ID splits and class-weighted training: 79.63% test accuracy, weighted F1 0.8075 and mean AUC 0.9522.

Personal Hand Rehabilitation Assistant: Dual-ESP32 master-slave glove with servo actuation.
Non-Invasive Bilirubin Detection: OPT101P photodiode with blue laser, red and green light.
Smart Pregnancy Pain-Relief Belt: ESP32 sensor-triggered vibration and thermal-control prototype.
IoT Environmental Monitoring for Hospitals: ESP32 sensing with Blynk monitoring and alerts.
Heart-Rate Monitoring Simulation: Arduino and LCD implementation in Proteus.
Temperature and Ultrasonic Proximity Simulation: LM35 and ultrasonic sensing in Proteus.
Microstrip Patch Antenna Analysis: MATLAB simulation of S11 and input impedance.

AI APPLICATIONS AND AUTOMATION

Mira Conversational AI Avatar: Built with Codex: React/TypeScript, Gemini Live, speech recognition and 3D lip sync.
n8n Automation and AI Projects
Weather Alert Bot: Weather API and webhook workflow.
AI Email Summaries and Telegram Notifications: Gemini email-subject summarization.

EXPERIENCE

Research Assistant | Biomedical Engineering Department, CUET Jul 2026–Present
Under Dr. Sumit Majumder; assigned patient-data collection at Chittagong Medical College and Hospital. Clinical collection/validation awaits ethical board approval.
Industrial Attachment | Parkview Hospital, Chittagong 7–27 May 2025
Three-week attachment covering hospital operations and medical-device familiarization.

CERTIFICATIONS AND LEADERSHIP

Coursera, 2026: n8n AI Agents & Secure APIs; AI Models/Prompt Engineering; Workflow Automation; Healthcare GenAI.
Vice President, Green for Peace Club, CUET; Project Development Secretary, CUET BMES; Treasurer, IEEE EMBS CUET.
Best Stall Award, CU Science Fair.